Analysis of Logit Lens on Visual Models

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POLITECHNIKA WARSZAWSKA

What is Logit Lens?

Logit Lens is an **interpretability technique** that projects intermediate layer activations of a neural network into the final output space — typically the model's **logits** (pre-softmax class predictions).

Literature Overview

[1] nostalgebraist, **Interpreting GPT: The Logit Lens.** LessWrong, 2022, URL: *https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens*

[2] Belrose et al., **Eliciting Latent Predictions from Transformers with the Tuned Lens**. arXiv preprint arXiv:2303.08112, 2023, URL: *https://arxiv.org/abs/2303.08112*

[3] Jiang et al., **Interpreting and Editing Vision-Language Representations to Mitigate Hallucinations**, *arXiv preprint arXiv:2410.02762*, 2024, URL: https://arxiv.org/abs/2410.02762, ICLR 2025 Poster

[4] Joseph et al., **Prisma: An Open Source Toolkit for Mechanistic Interpretability in Vision and Video**, arXiv preprint arXiv:2504.19475, 2025, URL: *https://arxiv.org/abs/2504.19475*

Motivations

Understanding how **internal representations** develop across a neural network's layers can reveal how **predictions evolve** from low-level features to high-level concepts.

How does the training method of visual transformers influence the formation of predictions and representations across layers?

Methodology

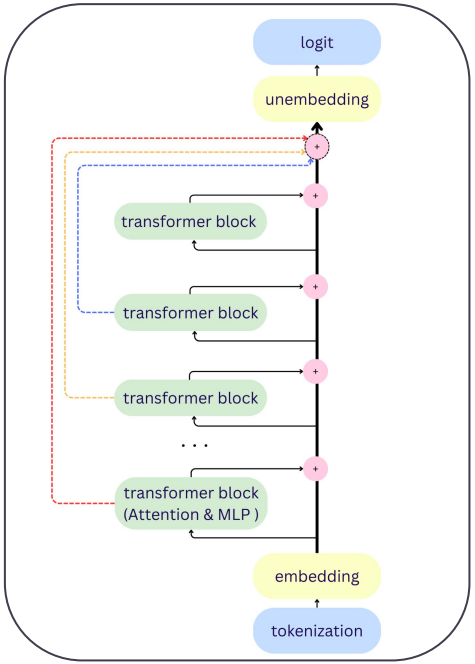
Dataset:

- Tiny ImageNet

Models:

- CLIP ViT-B/32 *(OpenAI)*
- CLIP ViT-L/14 *(OpenAI)*
- DINO *(timm)*

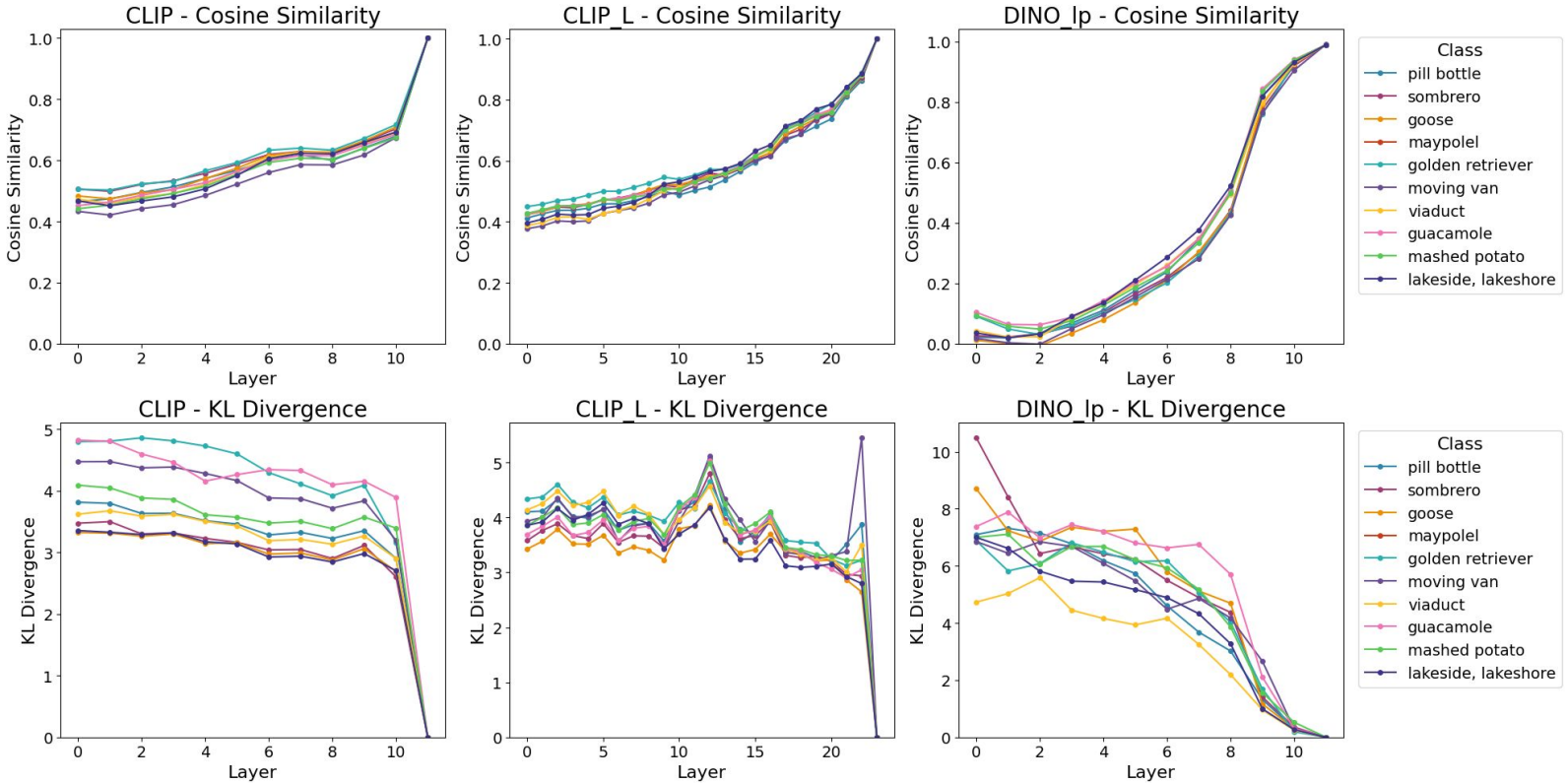
Logit Lens:



Metrics and measures:

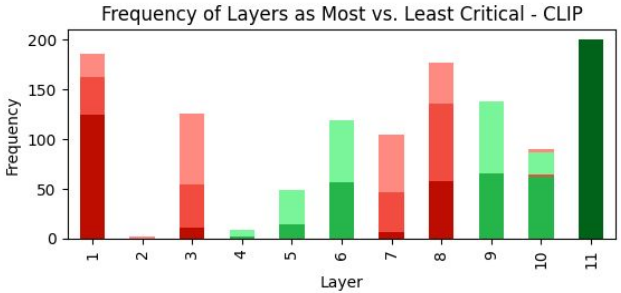
- cosine similarity
- KL divergence

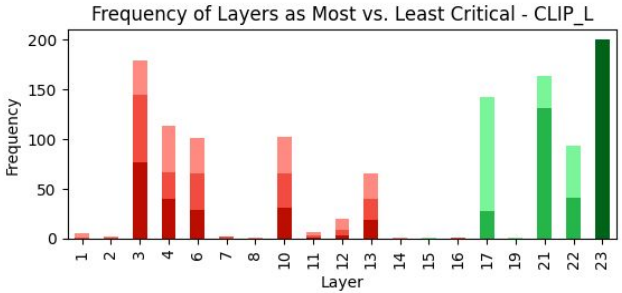
Results – representation similarities

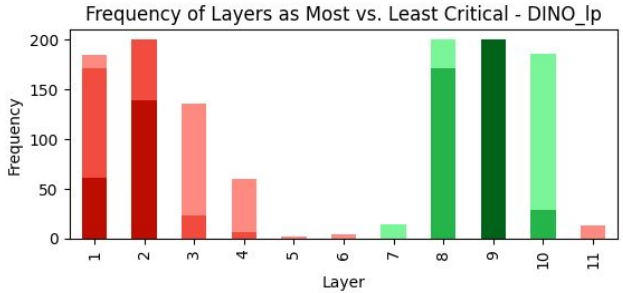


Results – critical layers

based on change in cosine similarity

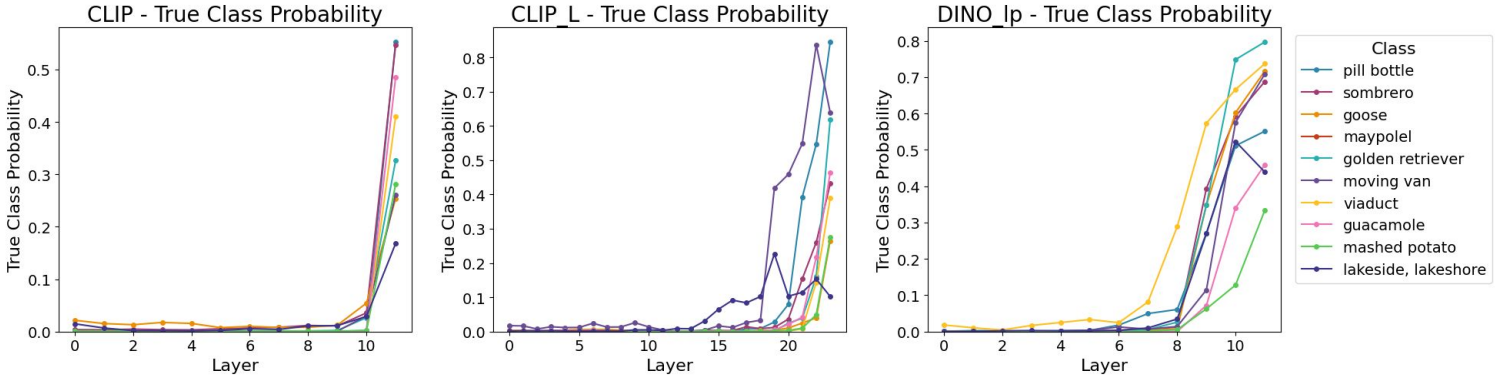


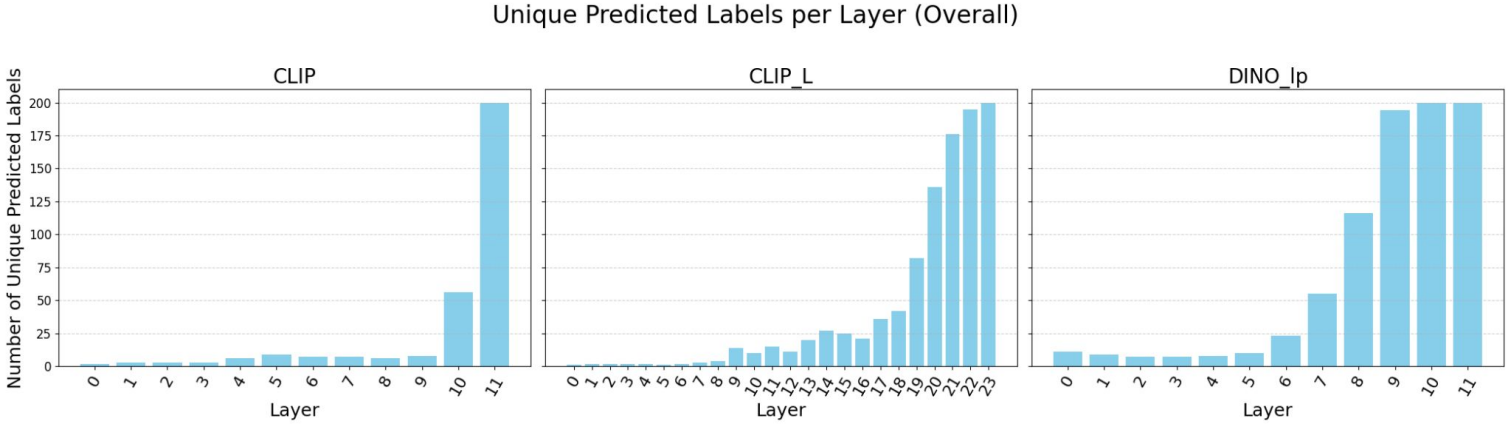




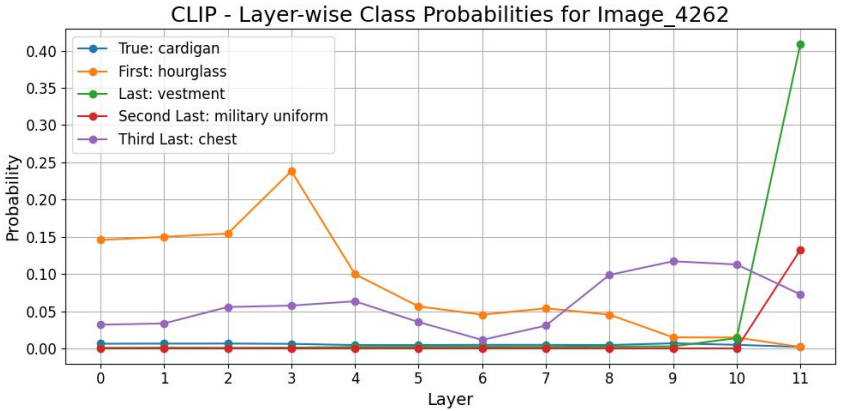


Results – true class probabilities



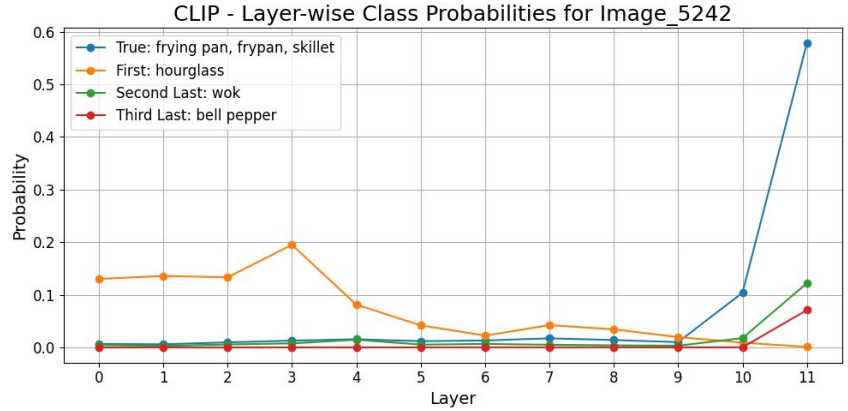


Results – prediction probabilities ViT-B





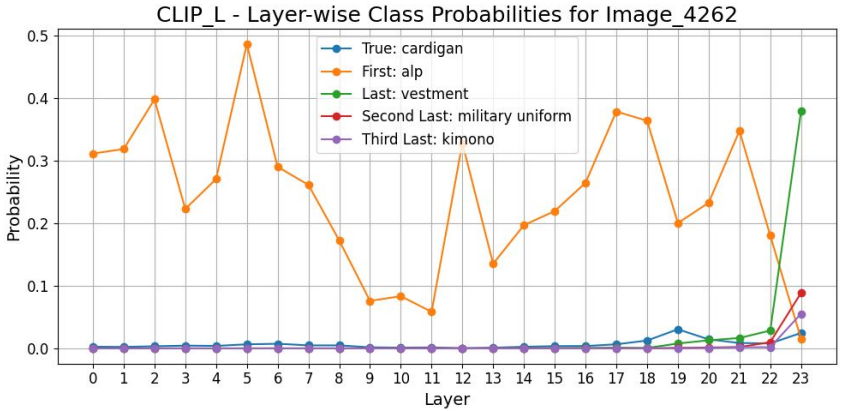
Final Predictions at Last Layer:
Predicted: **vestment** (0.4087)
Second most likely: **military uniform** (0.1324)
Third most likely: **chest** (0.0726)





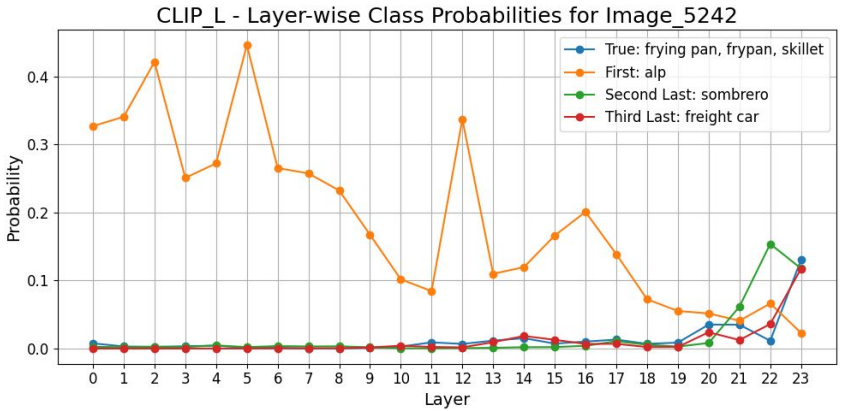
Final Predictions at Last Layer:
Predicted: **frying pan, frypan, skillet** (0.5779)
Second most likely: **wok** (0.1220)
Third most likely: **bell pepper** (0.0709)

Results – prediction probabilities ViT-L





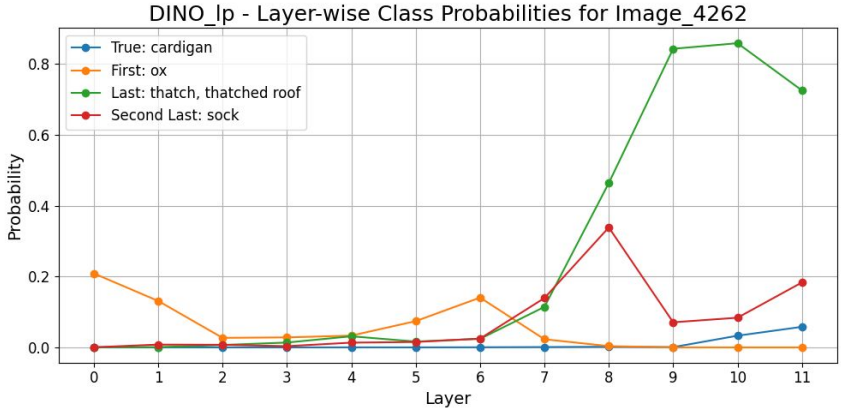
Final Predictions at Last Layer:
Predicted: **vestment** (0.3791)
Second most likely: **military uniform** (0.0887)
Third most likely: **kimono**(0.0554)





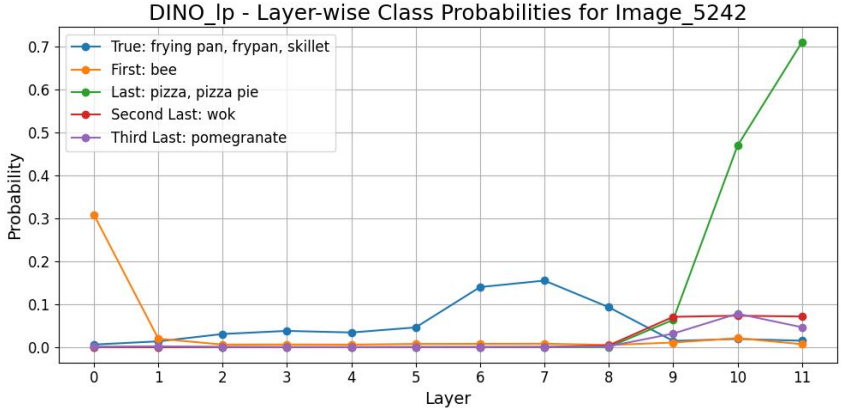
Final Predictions at Last Layer:
Predicted: **frying pan, frypan, skillet** (0.1306)
Second most likely: **sombrero** (0.1177)
Third most likely: **freight car** (0.1175)

Results – prediction probabilities DINO





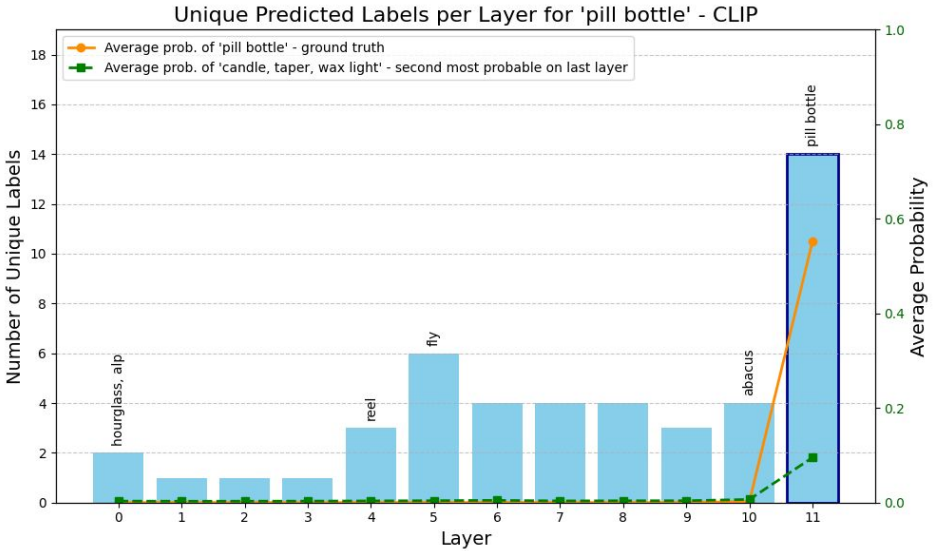
Final Predictions at Last Layer:
Predicted: **thatch, thatched roof** (0.7255)
Second most likely: **sock** (0.1833)
Third most likely: **cardigan** (0.0581)

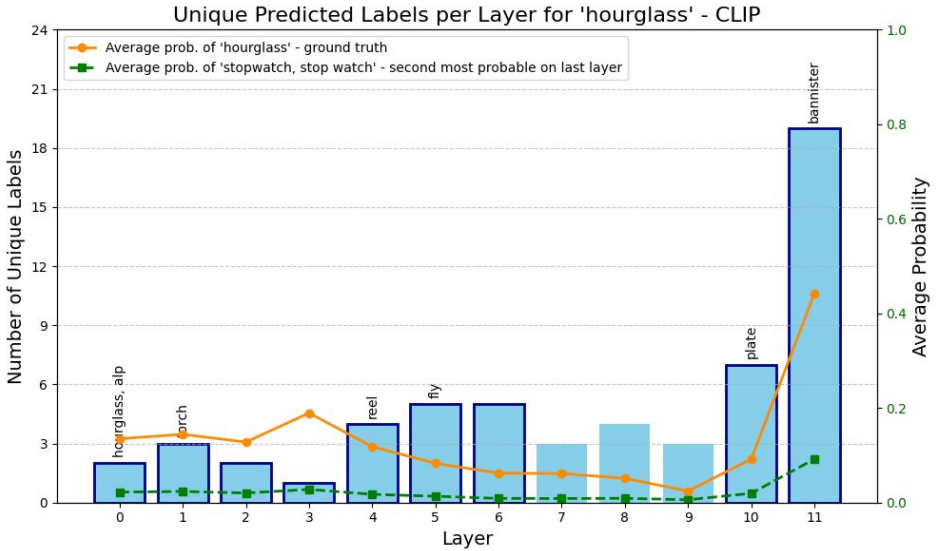




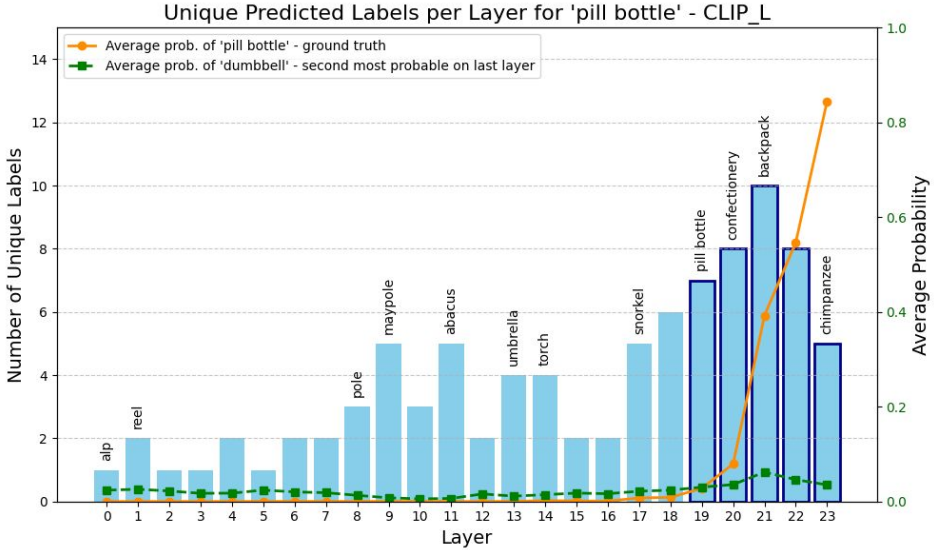
Final Predictions at Last Layer:
Predicted: **pizza, pizza pie** (0.7095)
Second most likely: **wok** (0.0715)
Third most likely: **pomegranate** (0.0466)

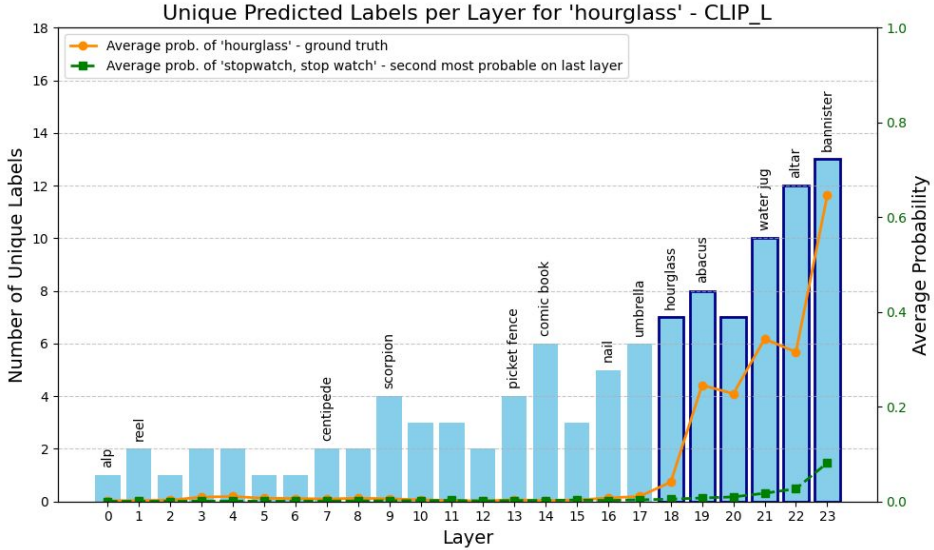
Results


Unique Predicted Labels per Layer for 'pill bottle' - CLIP
Average prob. of 'pill bottle' - ground truth
Average prob. of 'candle, taper, wax light' - second most probable on last layer
Number of Unique Labels
Average Probability
Layer
hourglass, alp
reel
fly
abacus
pill bottle

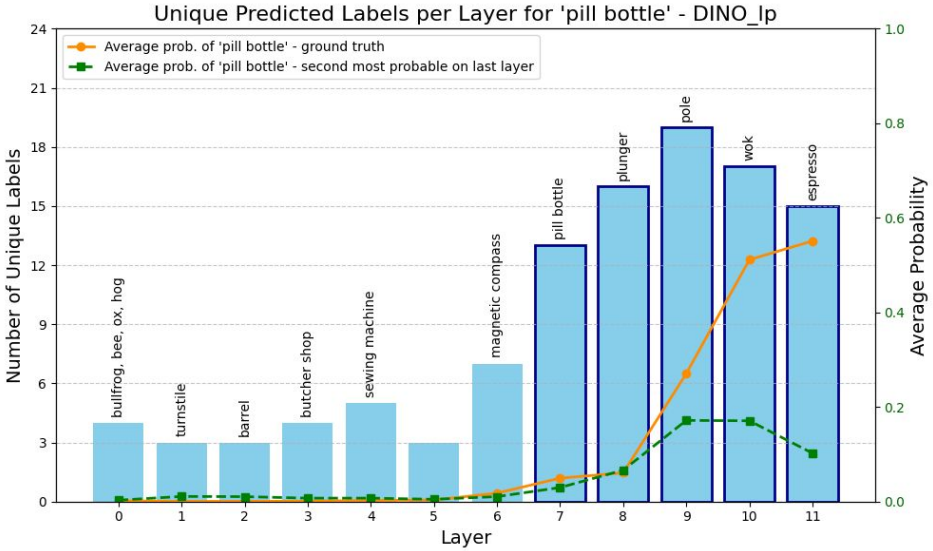

Unique Predicted Labels per Layer for 'hourglass' - CLIP
Average prob. of 'hourglass' - ground truth
Average prob. of 'stopwatch, stop watch' - second most probable on last layer
Number of Unique Labels
Average Probability
Layer
hourglass, alp
orch
reel
fly
plate
bannister

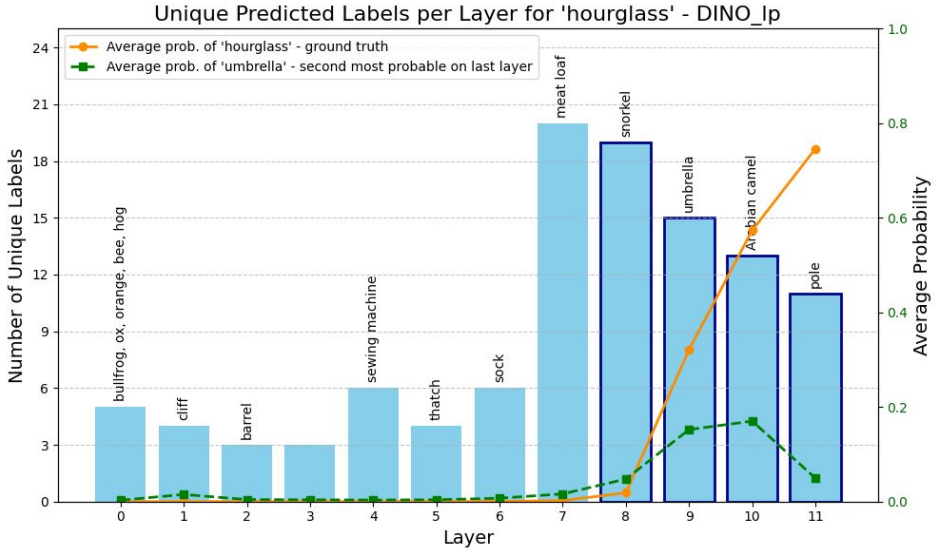
Results


Unique Predicted Labels per Layer for 'pill bottle' - CLIP_L
Average prob. of 'pill bottle' - ground truth
Average prob. of 'dumbbell' - second most probable on last layer
Number of Unique Labels
Average Probability
Layer
alp
reel
pole
maypole
abacus
umbrella
torch
snorkel
pill bottle
confectionery
backpack
chimpanzee


Unique Predicted Labels per Layer for 'hourglass' - CLIP_L
Average prob. of 'hourglass' - ground truth
Average prob. of 'stopwatch, stop watch' - second most probable on last layer
Number of Unique Labels
Average Probability
Layer
alp
reel
centipede
scorpion
picket fence
comic book
nail
umbrella
hourglass
abacus
water jug
altar
bannister

Results


Unique Predicted Labels per Layer for 'pill bottle' - DINO_lp
Average prob. of 'pill bottle' - ground truth
Average prob. of 'pill bottle' - second most probable on last layer
Number of Unique Labels
Average Probability
Layer
bullfrog, bee, ox, hog
turnstile
barrel
butcher shop
sewing machine
magnetic compass
pill bottle
plunger
pole
wok
espresso


Unique Predicted Labels per Layer for 'hourglass' - DINO_lp
Average prob. of 'hourglass' - ground truth
Average prob. of 'umbrella' - second most probable on last layer
Number of Unique Labels
Average Probability
Layer
bullfrog, ox, orange, bee, hog
cliff
barrel
sewing machine
thatch
sock
meat loaf
snorkel
umbrella
Arabian camel
pole

Contributions Summary

This work deepens understanding of how transformers behave at different stages of learning by comparing multimodal (CLIP) and self-supervised (DINO) models. We show how different training objectives influence the way these models build predictions layer by layer. Our analysis provides insights into when and how class-related information emerges throughout the network and introduces a method for pinpointing the most critical layers in each of the models.